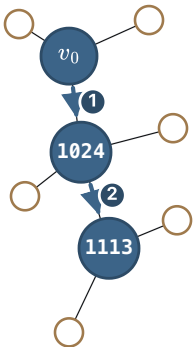


(a) Graph walk trajectory

the LLM walks the graph, picking neighbours



user
classify $v_0 \rightarrow c \in \mathcal{Y}$

graph env

$x, \mathcal{N}(v_0) = \{1024, 1450, \dots\}$

π_θ {<thinking>...</thinking>}

<walk>1024</walk>

graph env

$x, \mathcal{N}(1024) = \{1113, 0905\}$

π_θ {<thinking>...</thinking>}

<walk>1113</walk>

π_θ {<thinking>...</thinking>}

<answer>c</answer>

(b) OPSD walk-action credit

does the recap $\bar{o}$ make the LLM's own pick more likely?



the LLM scores its own pick twice

$\bar{o}$ is the walk's own recap: no extra rollouts, no step labels

student

blind

$s_{<t} \xrightarrow{\pi_\theta} \pi_t^{\text{stu}}(X)$

50%

teacher

with $\bar{o}$

$s_{<t} \oplus \bar{o} \xrightarrow{\pi_\theta} \pi_t^{\text{tea}}(X)$

72%

$$\text{KL}_t^{\text{rev}} = \log \pi_t^{\text{stu}} - \log \pi_t^{\text{tea}} \approx -0.36$$

the same per-step credit applies at every t

reward $R(\tau)$

format · answer · action

GRPO group-relative

$$\hat{A}_\tau = (R - \mu_R) / \sigma_R$$

OPSD advantage

$$\tilde{A}_t = \hat{A}_\tau - \beta m_t \text{KL}_t^{\text{rev}}$$

→ to PPO

LLM π_θ

graph env & observation

OPSD advantage